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Channel length effects on the performance of vOECTs

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Abstract

Channel length (L), the most important parameter of various transistors, determines the transistor performance and integration density. Emerging vertical organic electrochemical transistors (vOECTs), easily enabling L of less than 100 nm, show extremely high transconductance (g_m) and small footprints. However, as a new device structure, systematic studies on the limit and effects of the L on transistor performances remain unexplored. Here, by adjusting the concentration of the organic mixed ionic-electronic conductor (OMIEC) channel solution, vOECTs with L from ~100 nm to ~15 nm are fabricated and characterized. Surprisingly, shorter L would not lead to higher transistor performance, where the highest peak g_m of ~183 mS is obtained with ~20 nm L. Through comparing the relationship between L, g_m, and response time, it was found that the influence of L on device performance decreases gradually when the L is reduced to below 50 nm. This is mainly due to the decreased ion injection capability resulting from the reduced channel thickness. This work demonstrates an optimal design of vOECTs channel geometry that will lead to state-of-the-art transistor performance and also provides insights into ionic-electronic coupling mechanisms in the nanoscale.

Introduction

Organic electrochemical transistors (OECTs) are promising building blocks for biosensors, wearable devices, and neuromorphic systems due to their extremely low power consumption [1–3], high transconductance (g_m) [4], and good biocompatibility [5]. Although great advancements have been made in OECTs, further improvements in g_m [6], response speed [7], and device size [8, 9] are continuously sought. Therefore, when aiming to improve the amplification or speed properties of the OECT, the channel geometry is the main factor to consider [10–12]. For conventional OECTs (cOECTs) with planar structures, reducing channel length (L) requires high-resolution photolithography and complex fabrication processes, which makes it challenging to downscale L to sub-micrometer regime [13–15]. Moreover, in cOECTs, enlarged channel thickness benefits high g_m but sacrifices response speed, due to longer ion diffusion distance [16–18].

Recently, an alternative approach to overcome the intrinsic trade-off limit between g_m and response speed has emerged by adopting vertical channel geometries [19–22], where sub-micrometer L becomes feasible. For example, Donahue *et al* achieved an L of 450 nm defined by the thickness of a parylene-C layer sandwiched between the source and drain electrodes [21]. By utilizing a porous top electrode, Yan *et al* demonstrated an L down to ~100 nm, exhibiting g_m up to 68.88 mS and a transient time (τ) of <300 μs [23]. Koutsouras *et al* fabricated a 60 nm vertical channel sandwiched between the source and drain electrodes by employing electrodeposition, giving rise to a high g_m of 275 mS [24]. Recently, our group and collaborators demonstrated a

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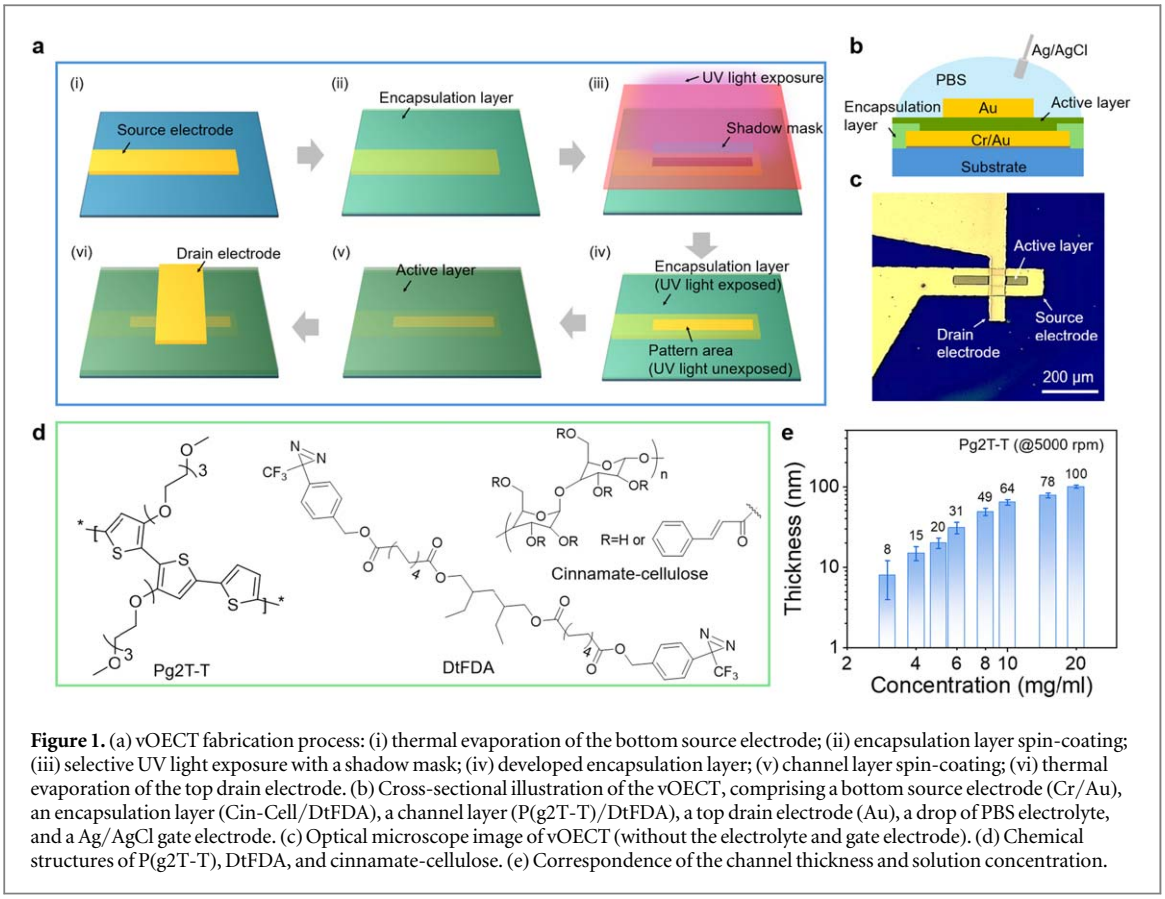


Figure 1. (a) vOECT fabrication process: (i) thermal evaporation of the bottom source electrode; (ii) encapsulation layer spin-coating; (iii) selective UV light exposure with a shadow mask; (iv) developed encapsulation layer; (v) channel layer spin-coating; (vi) thermal evaporation of the top drain electrode. (b) Cross-sectional illustration of the vOECT, comprising a bottom source electrode (Cr/Au), an encapsulation layer (Cin-Cell/DtFDA), a channel layer (P(g2T-T)/DtFDA), a top drain electrode (Au), a drop of PBS electrolyte, and a Ag/AgCl gate electrode. (c) Optical microscope image of vOECT (without the electrolyte and gate electrode). (d) Chemical structures of P(g2T-T), DtFDA, and cinnamate-cellulose. (e) Correspondence of the channel thickness and solution concentration.

vertical channel with L of ~ 100 nm, showing ultra-high g_m (377.6 μS μm^{-2}) and excellent cycling stability [25]. It is worth noting that, in vertical OECTs (vOECTs), both bottom and top electrode widths would affect the transistor performance tremendously, where wider electrode width would typically lead to higher current and slower switching speed. Detailed correlations of the electrode width with transistor performances can be found in previous research [19, 26]. Despite the success of vOECTs with short L in sub-micrometer ranges, the impact of L on the performance of vOECTs has not yet been fully understood. Moreover, due to the immature device model of vOECT, how the key parameters (including g_m and τ) would vary upon ultra-short L, and how short the L can be in vOECTs, remain unclear.

Here, by modulating the concentration of the organic mixed ionic-electronic conductor (OMIEC) channel solutions, precisely controllable channel thickness, viz. L of vOECT in the range of 15 ~ 100 nm, is realized. It is revealed that an optimal L exists, beyond which further reduction does not significantly enhance device performance. Specifically, vOECTs with L of ~20 nm show excellent electrical performance with an on-state current of up to 25 mA, a fast transient response of τ_{on} and τ_{off} of 2.27 ± 0.12 and 0.62 ± 0.05 ms, and a high g_m of ~180 mS at a drain voltage (V_D) of −0.1 V. More importantly, the regulatory mechanism of nanoscale L on key OECT parameters such as g_m and τ is investigated. By comparing device performance with varying L, it is discovered that when L decreases to below 50 nm, the enhancement of g_m and transient response time gradually weakens, suggesting that the influence of L on device performance becomes less significant under $L < 50$ nm. This work not only identifies the optimal design parameters for vOECTs but also highlights the pivotal role of L in determining device performance, offering valuable guidance for developing next-generation vOECTs with enhanced characteristics.

Results and discussion

The schematic diagram and fabrication process of the vOECTs are presented in figures 1(a)–(c), where the detailed fabrication process is presented in the Experimental section. The bottom and top Au electrodes serve as the source and drain electrodes, respectively. The encapsulation layer, composed of cinnamate-cellulose and trifluoromethyl diazirine (DtFDA) in a weight ratio of 4:1, features a 35 × 250 μm rectangular opening right on top of the bottom electrode (figures 1(b) and S1). Here, such an encapsulation layer can effectively prevent short circuits caused by the direct contact of the top and bottom electrodes, as thin channel thickness may lead to direct exposure of the bottom electrode edge (figure S2). A transistor channel based on a mixture of a p-type

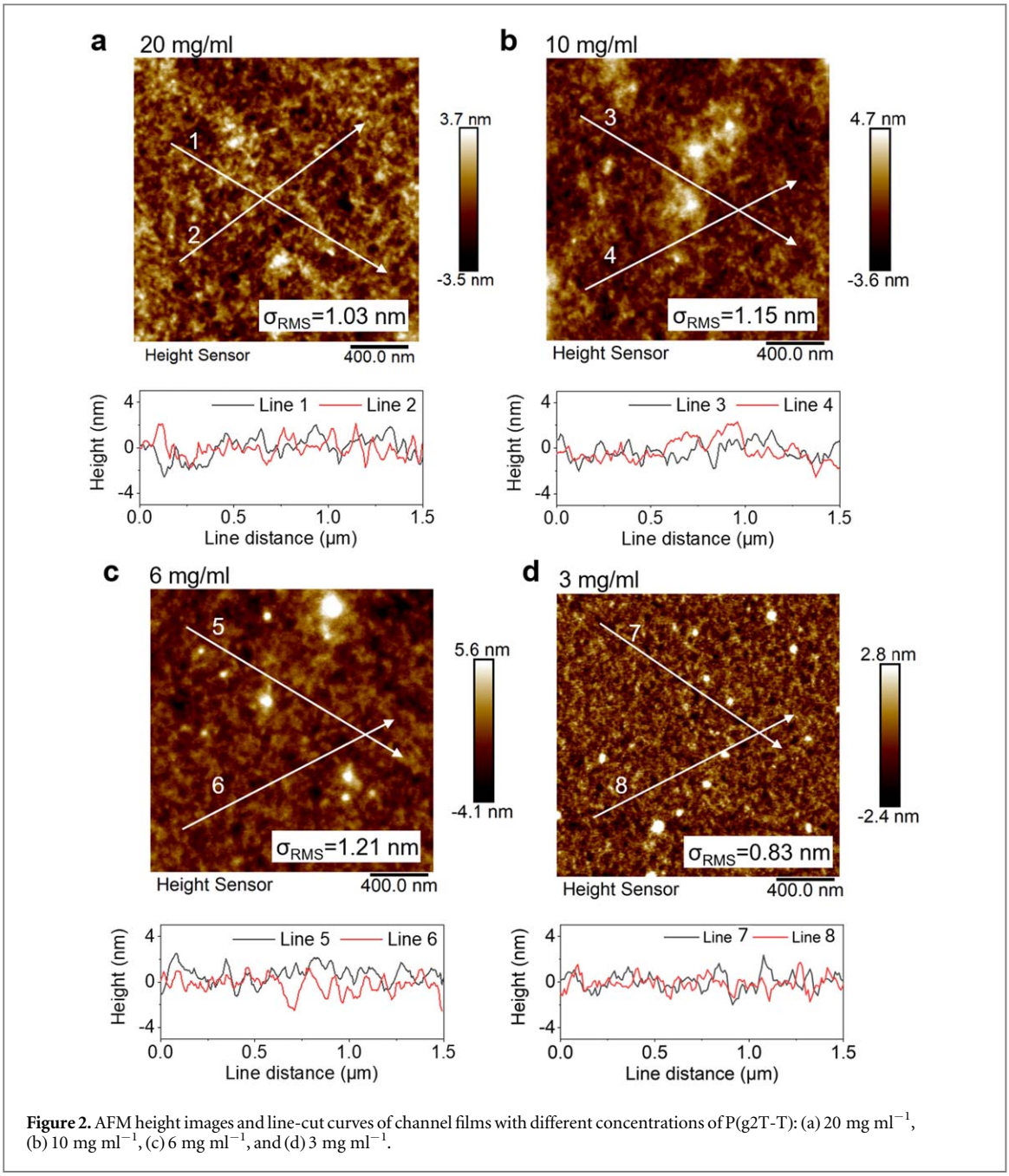


Figure 2. AFM height images and line-cut curves of channel films with different concentrations of P(g2T-T): (a) 20 mg ml^{-1}, (b) 10 mg ml^{-1}, (c) 6 mg ml^{-1}, and (d) 3 mg ml^{-1}.

OMIEC [poly(2-(3,3′-bis(2-(2-(2-methoxyethoxy)ethoxy)ethoxy)-[2,2′-bithiophen]-5)yl thiophene) (P(g2T-T))] and a photo-crosslinker [double-end trifluoromethyl diazirines (DtFDA)] (weight ratio = 1:1) was formed with its thickness controlled by the concentration of the P(g2T-T)/DtFDA blend solution (*vide infra*). The final vOECT structure exhibited a channel area of 35 × 40 μm (figure 1(c)) given by the overlapping area of the bottom and top electrodes. The chemical structures of P(g2T-T), DtFDA, and cinnamate-cellulose are shown in figure 1(d). Specifically, to control the channel thickness (viz. *L* of vOECT), the concentrations of P(g2T-T): DtFDA solution varied from 3 to 20 mg ml^{-1}. As shown in figure 1(e), with increasing solution concentration, the channel thickness gradually increases from 8 ± 4 nm to 100 ± 4 nm.

Next, atomic force microscopy (AFM) was applied to examine the surface morphologies of the channel layers with different thicknesses. As shown in figure 2 and figure S3, all films with various thicknesses exhibit an interconnected and fibrillar morphology in AFM height images. Meanwhile, variations in film thickness show a minor effect on the surface roughness. As shown in figures 2(a)–(c) and S3, all films exhibit a relatively smooth surface, where the surface root-mean-square roughness (σ_{RMS}) remains below 1.5 nm in all film thickness ranges of 8 ~ 100 nm. From the line cuts of the AFMs, it is obvious that all films show less than 6 nm variations in film thickness. However, note that in films with ~8 nm based on 3 mg ml^{-1} precursor solution, such a small variation could lead to severe current leakage as the thinnest film part could be less than 2 nm.

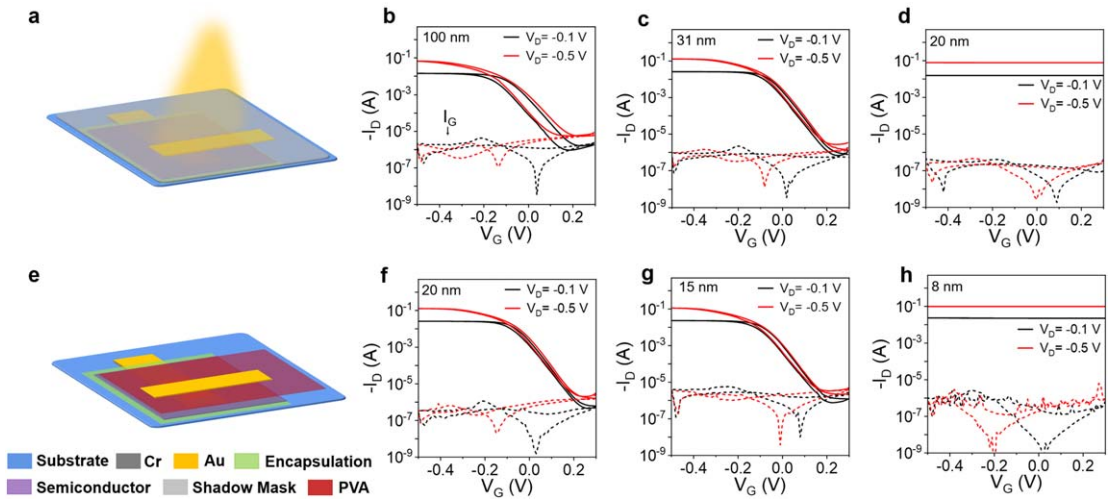


Figure 3. Transfer characteristics of vOECTs with (a) thermal evaporated top electrodes, where corresponding channel lengths are: (b) 100 nm, (c) 31 nm, and (d) 20 nm. (e) Transfer-printed top electrodes with corresponding channel lengths: (f) 20 nm, (g) 15 nm, and (h) 8 nm.

Subsequently, transistor performances are characterized on these vOECTs with different L. As L decreases from 100 nm to 31 nm, a noticeable alleviation in the hysteresis of the transfer characteristics is observed (figures 3(b)–(c) and S4). However, when the L is further reduced to 20 nm, no modulation on the drain current (I_D) by gate voltage (V_G) is observed, where I_D remains at 15.8 mA ($V_D = -0.1$ V) or 78.1 mA ($V_D = -0.5$ V) (figure 3(d). This is mainly attributed to the penetration of the hot Au atoms into the channel during thermal evaporation, resulting in direct contact with the top and bottom electrodes [27]. To avoid the damage caused by thermal evaporation to films thinner than 20 nm, a transfer printing process for the fabrication of the top electrode is then adopted. As shown in figure S5, such a process involves polyvinyl alcohol (PVA) sacrificing layer deposition, gold evaporation on the PVA layer, PVA peeling off, Au electrode transfer, and redissolving PVA (Detailed fabrication processes can be found in the Experimental section). To clarify the impact of direct evaporation and transfer printing on the performance of vOECT, device performances of 100 nm L-based vOECTs fabricated by these two methods are characterized, respectively. As shown in figure S6 and table S1, these two devices show similar performances with on current (I_{on}, I_D under $V_D = -0.5$ V and $V_G = -0.5$ V) of 61.5 ± 1.7 mA and 51.3 ± 4.1 mA, and g_m of 200.67 ± 13.67 mS and 195.32 ± 14.79 mS, for the vOECTs based on evaporated and transfer printed top electrodes, respectively. Therefore, the use of direct evaporation and transfer printing methods has a minimal impact on device performance. Thus, for vOECTs with a 20 nm and 15 nm thick semiconductor layer, the top drain electrode fabrication uses the transfer printing method (figure 3(e). However, when the L is reduced to 8 nm, the device still exhibits short-circuit behavior. As shown in the AFM (figure S3), due to the surface roughness, the film based on 3 mg ml^{-1} precursor may exhibit a thinnest film thickness of only ~2 nm, which could easily lead to quantum tunneling, ultimately causing a short circuit (figure 3(h).

Representative transfer characteristics and electrical performances of vOECT with various channel lengths are summarized in figure 4. As shown in figure 4(a), transistor performances are affected tremendously by L. vOECTs with 100 nm L exhibit the lowest I_{on} of 13.8 ± 0.4 mA with obviously I-V hysteresis. As L further decreases, I_{on} first increases to 18.0 ± 1.2 mA ($L = 78$ nm) and 25.2 ± 0.9 mA ($L = 20$ nm), but then slightly decreases to 22.9 ± 1.9 mA ($L = 15$ nm). Concurrently, the I_{off} exhibits a smooth trend: it starts at $(10.34 \pm 1.27) \times 10^{-7}$ A for 100 nm, significantly decreases to $(6.20 \pm 1.63) \times 10^{-7}$ A at 78 nm, and then fluctuates with minor increases and decreases as the L continues to shorten, reaching $(6.79 \pm 3.06) \times 10^{-7}$ A at 15 nm (figure 4(b). Similar to the trend observed in I_{on}, peak g_m exhibits a significant increase as the L decreases from 100 nm to 20 nm. Specifically, the peak g_m increases from 75.31 ± 5.51 mS at 100 nm, to 183.44 ± 5.79 mS at 20 nm. However, as the L is further reduced to 15 nm, the peak g_m slightly decreases to 150.43 ± 11.47 mS (figure 4(c), which can be linked to the reduction in I_{on}. Therefore, as L decreases from 100 nm to 31 nm, the current on/off ratio (I_{on}/I_{off}) significantly increases from $(1.35 \pm 0.17) \times 10^4$ to $(4.15 \pm 0.74) \times 10^4$ due to enhanced I_{on}. While, as L decreases to 20 and 15 nm, I_{on}/I_{off} slightly decreases to $(4.01 \pm 1.32) \times 10^4$ and $(3.91 \pm 1.72) \times 10^4$, respectively, due to lower I_{on}. Additionally, turn-on voltage (V_{on}) increases from 0.19 ± 0.02 V at 100 nm to 0.27 ± 0.01 V at 15 nm (figure 4(d), revealing that shorter L contributes to easier charge carrier injection due to drain-induced barrier lowering effect.

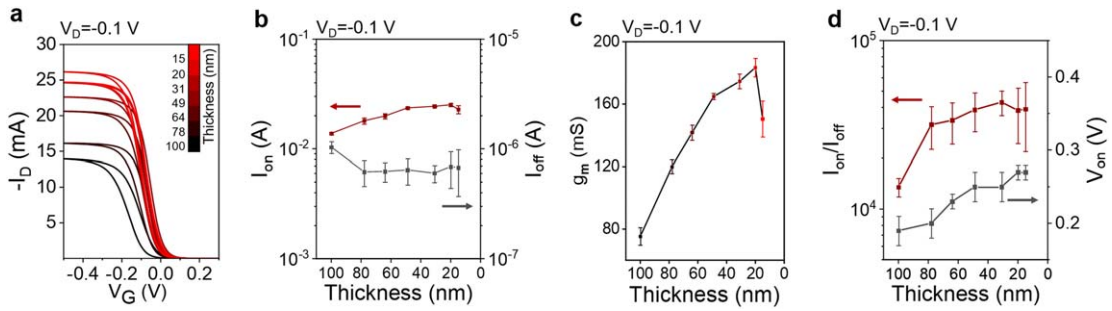


Figure 4. Representative (a) transfer characteristics and (b) variation of I_{on} and I_{off} in vOECT concerning different channel lengths. (c) Corresponding g_m of vOECT with different channel lengths. (d) Dependence of on/off ratio on the L from 100 to 15 nm.

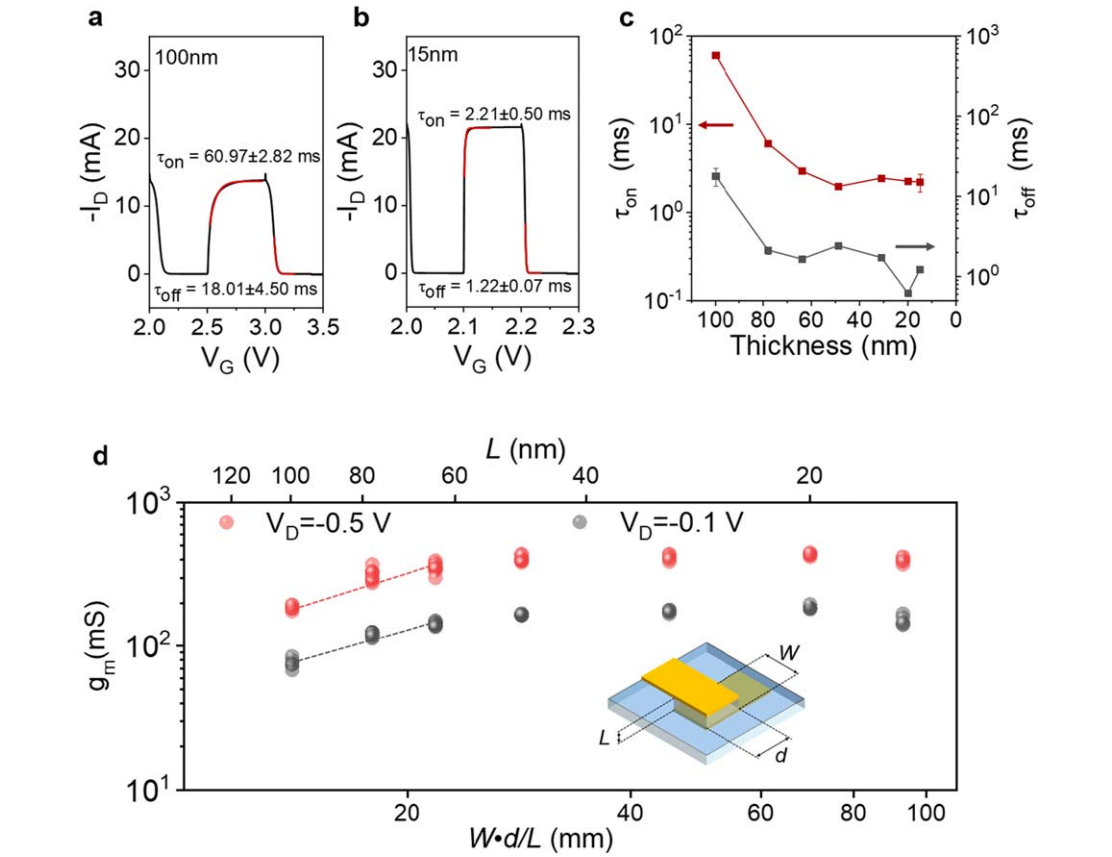


Figure 5. Transient response of vOECT with (a) 100 nm and (b) 15 nm of L. (c) Dependence of transient time on L of vOECTs. (d) Scaling of vOECT g_m at saturation for devices of varying geometry. Inset: vOECT configuration and channel dimension.

In addition, transient responses of the vOECTs with different L are further analyzed. As shown in figures 5(a)–(b) and figure S7, it is found that turn-on (τ_{on})/turn-off (τ_{off}) transient times decrease from 60.97 ± 2.82/18.01 ± 4.50 ms to 2.21 ± 0.50/1.22 ± 0.07 ms as L decreases from 100 nm to 15 nm, indicating that a faster ionic diffusion velocity is obtained in shorter L due to enhanced electric field. On the other hand, it is worth noting that τ would not always decrease with shorter L. As shown in figure 5(c), both τ_{on} and τ_{off} decrease significantly from 60.97 ± 2.82 and 18.01 ± 4.50 ms to 1.97 ± 0.09 and 2.43 ± 0.05 ms when L reduces from 100 to 49 nm. However, both τ remain relatively stable after L is <49 nm. Such a phenomenon indicates that ionic velocity would not be able to be further enhanced with a shorter channel as the electric field in the channel is speculated to saturate after L is shorter than ~50 nm. To further shorten τ, precise patterning of the channel layer and encapsulation of the electrodes, along with well-defined electrolyte distribution and gate electrode, are expected to enable OECTs with faster transient behaviors [28–30].

As shown in figure 5(d), (a) vOECT channel geometry configuration indicates that the L is the semiconductor layer thickness, and the widths of the bottom and the top electrodes define the channel

width (W) and the nominal depth (d) of the semiconductor, respectively. To better understand the influence between channel geometry and g_m, the notion of volumetric capacitance (C^*) in OECTs was introduced in a model describing transistor operation. Therefore, g_m is given by [12]:

$$g_m = \frac{Wd}{L}\mu C^*(V_T - V_G) \quad (1)$$

where μ is the hole mobility in the OSC channel and V_T is the threshold voltage. This equation explicitly shows the scaling of g_m with $\frac{Wd}{L}$. Thus, in cOECTs, g_m is highly dependent on the channel geometry, where g_m usually demonstrates linear behavior to $\frac{Wd}{L}$ [23].

As depicted in figure 5(d), for devices with channel lengths of 100, 78, and 64 nm, g_m indeed shows a linear correlation against $\frac{Wd}{L}$. However, when L is less than 64 nm, g_m does not increase with increasing $\frac{Wd}{L}$. This observation indicates a completely different result compared to that in cOECTs. The main reason for the difference is that in cOECTs, d (the channel depth), which corresponds to the thickness of the OSC film (typically in nanometers), is much smaller compared to W and L (typically in micrometers). As a result, $\frac{Wd}{L}$ is usually small in cOECTs (typically in micrometers). However, for vOECTs, the values of $\frac{Wd}{L}$ are significantly higher, reaching the millimeter scale, which is three orders of magnitude higher than cOECT. Such a large geometry factor could exaggerate other side effects. For instance, parasitic resistances, such as interface resistance or electrode series resistance, become particularly severe when the channel resistance is low, as in the case of vOECTs. Consequently, under on-state, the main factor that limits I_D and g_m would be the series resistances from the source/drain electrodes and electrode/channel interfaces. As demonstrated in figure S8, with shorter L, output characteristics under high V_G show no saturation region but only a linear dependence on V_D, reflecting that channel resistance is much smaller than series resistances [31]. Such observation becomes more obvious when L is reduced to ⩽49 nm. Note that a detailed analysis of such a phenomenon is still under investigation.

Conclusion

In this work, a systematic investigation into the impact of L on the performance of vOECTs is conducted, revealing that reducing L does not necessarily enhance transistor performance. Instead, an optimal L of ∼20 nm in P(g2T-T)-based vOECTs is identified, which results in the highest peak g_m of approximately 183 mS. Reduction in L to below 50 nm results in less response in device performance. As L decreases, g_m significantly increases and reaches a peak at L around 20 nm, after which it slightly decreases. This suggests that below a certain L threshold, the performance gains diminish, potentially due to the increased dominance of series resistances from the source/drain electrodes and electrode/channel interfaces, along with limited space and materials for the channel to swell. Additionally, our research also highlights that reducing L from 100 nm to 15 nm significantly improves switch speed of vOECTs due to increased electric field strength, accelerated ionic diffusion and smaller C^*. This work not only identifies an optimal design for vOECTs channel geometry for achieving superior transistor performance but also sheds light on the ionic-electronic coupling mechanisms at the nanoscale.

Experimental section

Materials

P(g2T-T) [32], DtFDA [33] and Cinnamate-Cellulose (Cin-Cell) [34] were synthesized based on previously reported methods. Isopropyl alcohol, acetone, and chloroform were purchased from Sigma-Aldrich. PVA was purchased from Aladdin. Micro-90 concentrated cleaning solution was obtained from International Products Corporation.

Solution preparation

P(g2T-T) was dissolved in chloroform with eight different concentrations (20 mg ml^{-1}, 15 mg ml^{-1}, 10 mg ml^{-1}, 8 mg ml^{-1}, 6 mg ml^{-1}, 5 mg ml^{-1}, 4 mg ml^{-1}, and 3 mg ml^{-1}). Cin-Cell and DtFDA were dissolved in chloroform with concentrations of 20 mg ml^{-1} and 100 mg ml^{-1}, respectively. These solutions underwent filtration through a 0.45 μm polyvinylidene difluoride filter. PVA was dissolved in water with a concentration of 10% under 90 °C. P(g2T-T) was mixed with DtFDA solution in a mass ratio of 1:1. Cin-Cell solution and DtFDA solution were mixed in a mass ratio of 4:1.

vOECTs fabrication with the thermal evaporated top electrode

vOECTs were fabricated on a pre-cleaned Si/100 nm SiO_2 wafer. First, 3 nm of Cr and 150 nm of Au were thermally evaporated with a shadow mask as the bottom source electrode. The substrates with the bottom

electrodes underwent a 15 min UV–ozone cleaning process. Then the Cin-Cell:DtFDA blend solution was spin-coated on the substrates at 3000 rpm for 20 s. The encapsulation layer was then patterned by exposure to 365 nm UV light (450 mW cm^{-2}) for 2 min with a shadow mask, followed by a 3 s development in chloroform and blow-dry. Then, the solution containing P(g2T-T):DtFDA mixture, was spin-coated onto substrates at 3000 rpm for 20 s. Subsequently, semiconductor films were crosslinked by exposure to 365 nm UV light (450 mW cm^{-2}) for 2 min. 150 nm of Au was then thermally evaporated with a shadow mask as the top drain electrode while maintaining the substrate at a temperature of 5 °C with a back water-cooling system. Finally, a droplet of phosphate buffer solution (PBS, 1×) was applied to the overlapping area of the electrode, and an Ag/AgCl electrode was inserted into the droplet to function as the gate electrode for the vOECT.

vOECTs fabrication via transfer printing

The procedures before the fabrication of the top electrode are the same as those for vOECT fabrication with a thermal evaporated top electrode. For the fabrication of the top electrode starts with spin-coating 10 wt% PVA aqueous solution onto a pre-cleaned glass substrate and annealing at 45 °C for 40 min. After annealing, the cured PVA film was removed from the glass substrate. Next, 150 nm of Au was evaporated onto the PVA film through a shadow mask. Then, the gold-coated PVA film is transferred and aligned on top of the target device. Subsequently, the gold-coated PVA film is softened and bonded to the target device by heating at 80 °C and pressing with a glass slide for 20 min. Later, the device was immersed in 60 °C deionized water for 240 min to dissolve the PVA film, resulting in top electrode deposition on the target device surface. Finally, a droplet of phosphate buffer solution (PBS, 1×) was applied to the overlapping area of the electrode, and an Ag/AgCl electrode was inserted into the droplet to function as the gate electrode for the vOECT.

vOECTs measurement and film characterization

Transfer and transient time characteristics of the vOECT were carried out with an FS-Pro (PDA) semiconductor parameter analyzer along with a probe station (Chengdu Chiptest Technology Co., Ltd) in ambient conditions. AFM characterizations were acquired with a Bruker Dimension Icon System in tapping mode under ambient conditions.

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Conflict of interest

There are no conflicts to declare.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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